

Cambridge IGCSE™

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0625/42

Practice Paper

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s²).

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages. Any blank pages are indicated.

- 1 Fig. 1.1 shows an electrically powered bicycle.

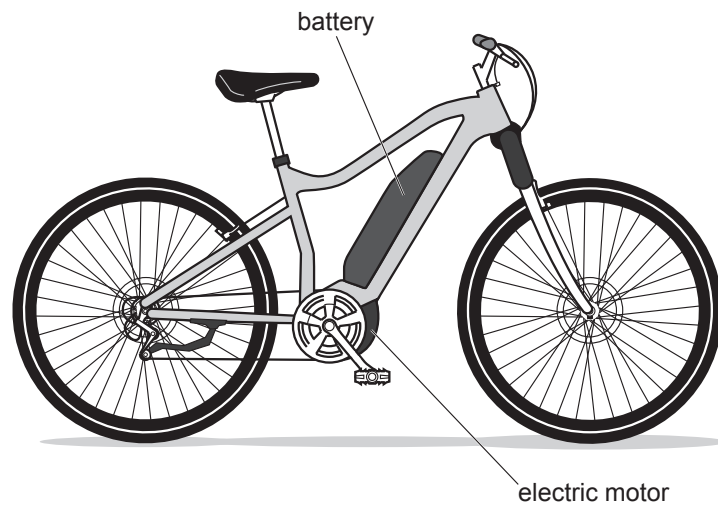


Fig. 1.1

The battery can deliver a power of 600 W for 60 min.

- (a) (i) Calculate the energy, in joules, stored in the battery when fully charged.

energy = J [3]

- (ii) State the form of energy stored by the battery.

..... [1]

- (b) The bicycle has a motor with an electrical input power of 250 W.

Calculate the time for which the battery can power the bicycle.

time = s [2]

- (c) State two environmental benefits of using an electrically powered bicycle instead of a small motorcycle.

1.

2.

[2]

[Total: 8]

- 2 Fig. 2.1 shows two objects connected by a cord passing over a pulley.

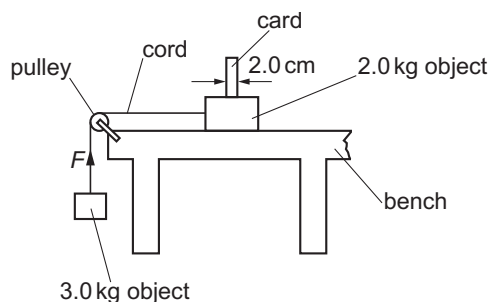


Fig. 2.1

The 2.0 kg object is released from rest and accelerates at 4.0 m/s^2 .

- (a) Calculate the resultant force acting on the 2.0 kg object.

force = N [2]

- (b) Calculate the upward force exerted by the cord on the 3.0 kg object.

force = N [3]

- (c) (i) The objects have a constant acceleration.

Show that the speed of the objects 0.80 s after release is 3.2 m/s.

[2]

- (ii) A card of width 2.0 cm passes through a beam of light at this speed. Calculate the time taken for the card to pass through the beam.

time = s [2]

[Total: 9]

- 3 (a) Fig. 3.1 shows water in a river moving parallel to the river bank at 4.0 m/s and a canoe travelling in the river.

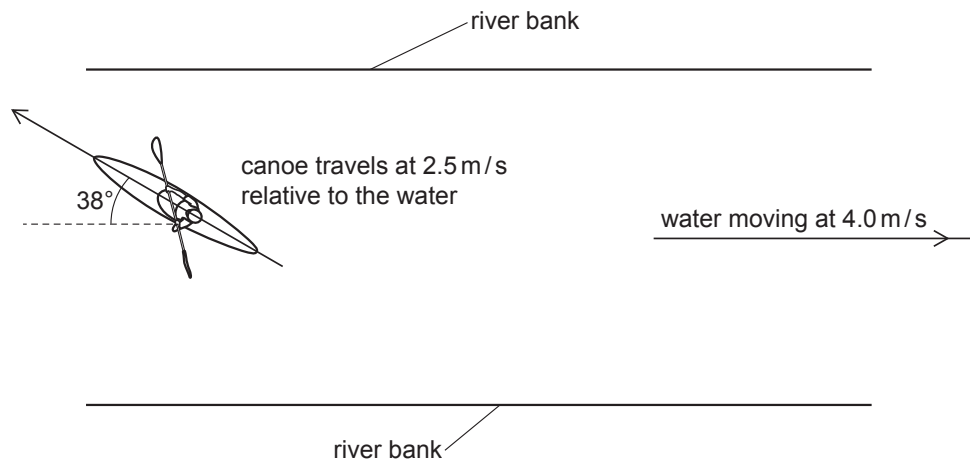


Fig. 3.1

The canoe travels at 2.5 m/s relative to the water and heads at an angle of 38° to the river bank. Draw a scale diagram to determine the canoe's resultant velocity and state the scale you used.

scale = [1]
 magnitude of resultant velocity = [2]
 direction of resultant velocity = [1]

- (b) Explain why the resultant velocity is not in the same direction as the canoe points.

.....
 [2]
 [Total: 8]

- 4 A student investigates the resistance of a wire.

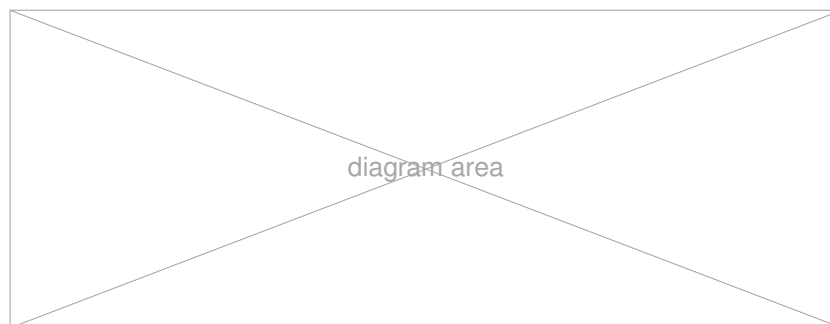


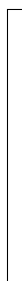
Fig. 4.1

- (a) Complete the table headings for the measurements needed in this experiment.

Length of wire / cm		
20		
40		
60		
80		

[2]

- (b) Describe how the student can use the measurements to determine the resistance of the wire.



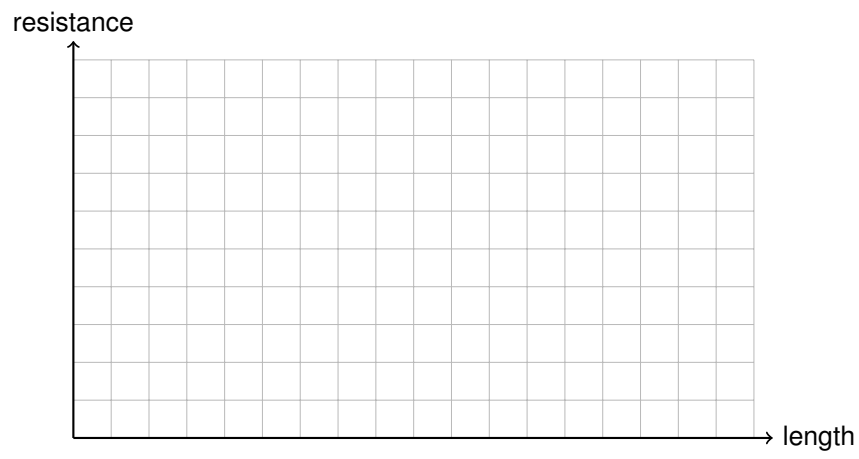
[4]

- (c) State two precautions that improve the reliability of the results.

1.
2.

[2]

- (d) On the grid, sketch the expected shape of a graph of resistance against length for a uniform wire.



[2]

[Total: 10]

BLANK PAGE

- 5 This page shows the structure for adding another long question. Replace this text with your own question stem.

(a) Write the first part of the question here.

answer = unit [2]

(b) Write the second part of the question here.

[4]
[Total: 6]